

Constant & Harmonic Perturbations.
(S.7.3) (S.7.4)

Oct-21

Constant Pert.

Given : $H_0 |n\rangle = E_n |n\rangle$

$$V(t) = \begin{cases} 0, & t < 0 \\ V, & t > 0 \end{cases}$$

Initial state $|a, t_0 = 0\rangle = |i\rangle$

transition amplitude to a particular state $|f\rangle$

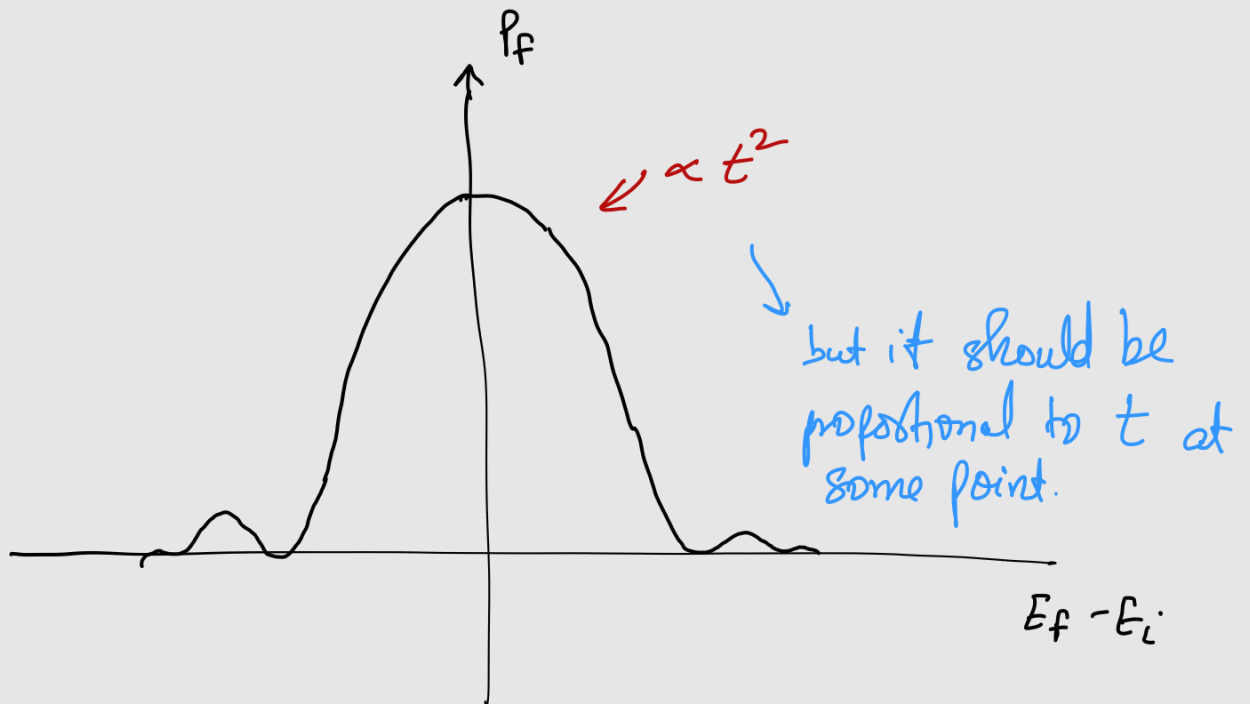
$$c_f^{(1)}(t) = \langle f | U_I^{(1)}(t, 0) | i \rangle = \frac{1}{i\hbar} \int_0^t dt_1 e^{i\omega_{fi}t_1} \underbrace{\langle f | V(t_1) | i \rangle}_{\text{const. } V_{fi}}$$

$$\omega_{fi} = \frac{E_f - E_i}{\hbar}$$

$$= \frac{1}{i\hbar} V_{fi} \frac{e^{i\omega_{fi}t} - 1}{i\omega_{fi}} = \frac{V_{fi}}{-\hbar\omega_{fi}} e^{i\omega_{fi}t/2} \underbrace{(e^{i\omega_{fi}t/2} - e^{-i\omega_{fi}t/2})}_{2i \sin(\omega_{fi}t/2)}$$

$$= \frac{2V_{fi}}{i\hbar\omega_{fi}} e^{i\omega_{fi}t/2} \sin\left(\frac{\omega_{fi}t}{2}\right)$$

$$\Rightarrow P(t) \propto |c_f^{(1)}(t)|^2 = \frac{4|V_{fi}|^2}{(E_f - E_i)^2} \sin^2\left(\frac{E_f - E_i}{2\hbar} t\right)$$



Total Transition Probability

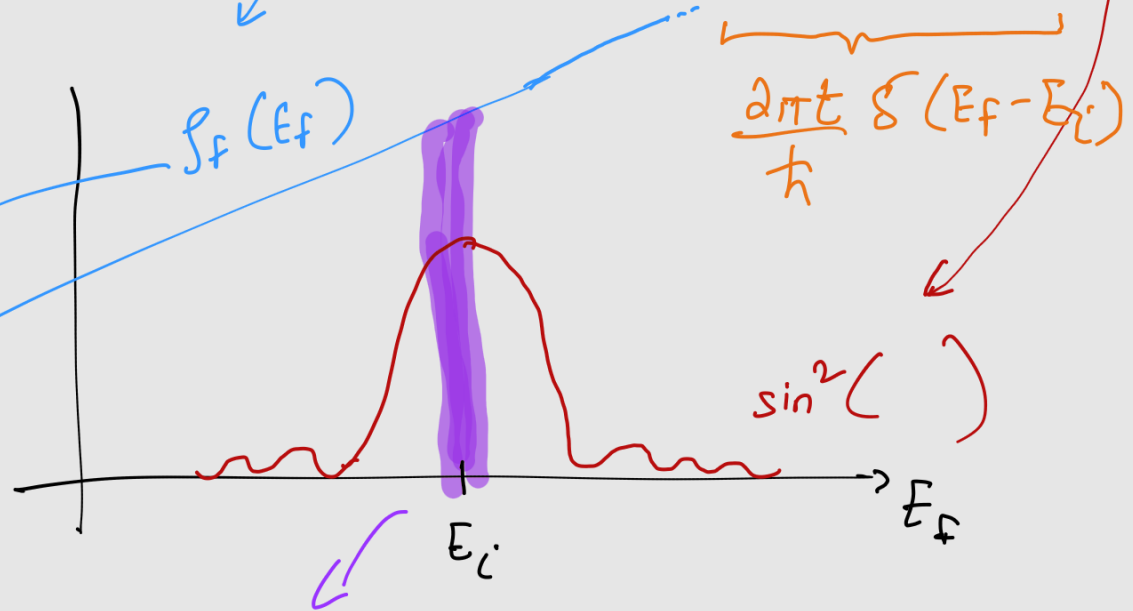
$F =$ set of possible final states $= \{f\}$

$$P_{i \rightarrow F}(t) = \sum_{f \in F} P_f(t) \longrightarrow \int dE_f \rho_f(E_f) P_f(t)$$

continuum
as the energy difference
gets smaller.

$$\frac{1}{T} \int dE_f \rho_f(E_f) |V_{fi}|^2 \frac{4 \sin^2\left(\frac{E_f - E_i}{2\hbar}\right)}{(E_f - E_i)^2}$$

some general
Pⁿ of energy



since the graph is narrower for $\sin^2()$ at E_i , the final Pⁿ (which is obtained by multiplying.

$\rho_f(E_f)$ & $\sin^2()$), the

main contribution would be from $E_f \rightarrow E_i$

$$\lim_{E_f \rightarrow E_i} \left(\rho_f(E_f) \sin^2\left(\frac{E_f - E_i}{2\hbar}\right) \right)$$

$$\frac{2\pi t}{\hbar} \delta(E_f - E_i)$$

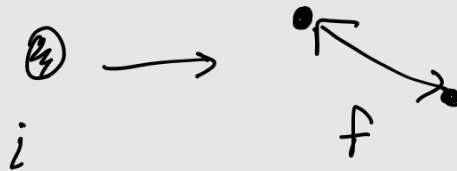
$$\downarrow \frac{2\pi}{\hbar} |V_{fi}|^2 \rho_f(E_f = E_i) t$$

Fermi's Golden Rule

$$\Gamma_{i \rightarrow F} = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho_f(E_f = E_i)$$

→ Applications

- spontaneous decay



- Scanning tunneling microscopy



Harmonic Perturbation

Example: Photoelectric effect

$$H(t) = H_0 + V(t)$$

Examples of $V(t)$:

$$V(t) \propto \cos(\omega t) = \frac{1}{2} e^{i\omega t} + \frac{1}{2} e^{-i\omega t}$$

$$V(t) \propto \sin(\omega t) = \frac{1}{2i} e^{i\omega t} - \frac{1}{2i} e^{-i\omega t}$$

General case:

$$V(t) = V e^{i\omega t} + V^\dagger e^{-i\omega t}$$

$$\left(\begin{array}{l} V^\dagger = V \\ \text{but } V^\dagger \neq V \end{array} \right)$$

Predict transitions between eigenstates of H_0

consider initial state $|i\rangle, E_i$

final state $|f\rangle, E_f$

transition amplitude:

$$c_f^{(1)}(t) = \frac{1}{i\hbar} \int_0^t e^{i\omega_{fi}t'} V_{fi}(t') dt', \quad \omega_{fi} = \frac{E_f - E_i}{\hbar}$$

$$= \frac{1}{i\hbar} \int_0^t e^{i\omega_{fi}t'} (V_{fi} e^{i\omega t'} + V_{fi}^* e^{-i\omega t'}) dt'$$

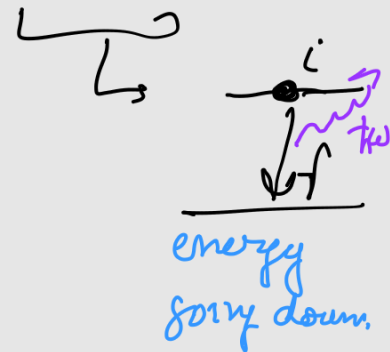
$$= \frac{1}{i\hbar} \left[\frac{e^{i(\omega_{fi} + \omega)t} - 1}{i(\omega_{fi} + \omega)} V_{fi} + \frac{e^{i(\omega_{fi} - \omega)t} - 1}{i(\omega_{fi} - \omega)} V_{fi}^* \right]$$

large when
 $\omega_{fi} = -\omega$

large when
 $\omega_{fi} = \omega$

- assume $\omega > 0$

first term : $\omega \approx -\omega_{fi} = \omega_{if} = \left(\frac{E_i - E_f}{\hbar} \right) > 0$

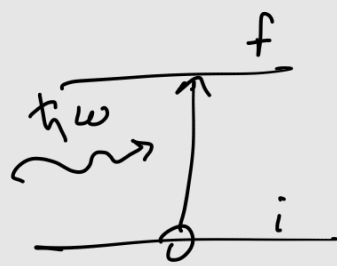


* transition most likely
for

$\hbar\omega \approx E_i - E_f$
photon energy

stimulated emission

Second term : $\omega \approx \omega_{fi} = \left(\frac{E_f - E_i}{\hbar} \right) > 0$



$$\hbar\omega \approx E_f - E_i$$

absorption.

Transition to a continuum

- first term : $\omega_f < \omega_i$

$$\langle f | U_I^{(1)}(t, 0) | i \rangle \approx \frac{1}{i\hbar} \frac{e^{i(\omega_{fi} + \omega)t} - 1}{i(\omega_f + \omega)} V_{fi}$$

$$P_f \approx 4 \left| \frac{V_{fi}}{\hbar} \right|^2 \frac{\sin^2((\omega_{fi} + \omega)t/2)}{(\omega_{fi} + \omega)^2}$$

sum over final states:

Emission:

$$\Gamma_{i \rightarrow [f]} = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho_f(E_f = E_i - \hbar\omega)$$

absorption:

$$\Gamma_{i \rightarrow [f]} = \frac{2\pi}{\hbar} |V_{fi}^+|^2 \rho_f(E_f = E_i + \hbar\omega)$$

